```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

```
PUBLIC_ENDPOINTS = {"auth.login", "static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
    ("用户与操作日志", "/users", "users"),
]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
```

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            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
      badges = {}
      if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
      def build(links):
            return [{"label": label, "href": href, "icon": icon,
                     "active": hit(href), "badge": badges.get(href, 0)}
                    for label, href, icon in links]
      groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
      ]
      #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
      if path == "/":
            page_title = "工作台"
      elif path.startswith("/profile"):
            page_title = "个人设置"
      elif path.startswith(("/users", "/logs")):
            page_title = "用户与操作日志"
      elif path.startswith("/search"):
            page_title = "搜索结果"
      else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                  if hit(href):
                        page_title = label
                        break
      #  层级引导用：项目是归档单位，图斑才是评价对象。
      #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
      from app.models import Project, MinePatch
      projects = Project.query.order_by(Project.id).all()
      return {
            "nav_groups": groups,
            "home_active": path == "/",
            "page_title": page_title,
            "today": date.today().strftime("%Y-%m-%d"),
            "search_q": request.args.get("q", ""),
            "role_labels": ROLE_LABELS,
            "project_count": len(projects),
            "patch_count": MinePatch.query.count(),
            "projects_without_patch": [p for p in projects
                                       if p.patches.count() == 0],
      }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
```

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    @app.after_request                                                  201
    def audit_writes(response):                                         202
        if request.method in ("POST", "PUT", "PATCH",                   203
    "DELETE"):                                                          204
            audit.record(request.endpoint,                              205
    response.status_code)                                               206
        return response                                                 207
    # 错误页：给出可读提示，而不是默认的英文错误页                      208
    from flask import render_template                                   209
    @app.errorhandler(403)                                              210
    def forbidden(e):                                                   211
        return render_template("error.html", code=403,                  212
                               message="没有权限执行该操作"), 403       213
    @app.errorhandler(404)                                              214
    def not_found(e):                                                   215
        return render_template("error.html", code=404,                  216
                               message="请求的页面不存在"), 404         217
    return app                                                          218
# ==== app/config.py ====                                             219
# -*- coding: utf-8 -*-                                               220
"""应用配置。"""                                                        221
import os                                                               222
from datetime import timedelta                                          223
BASE_DIR = os.path.abspath(os.path.dirname(__file__))                   224
class Config:                                                           225
    SECRET_KEY = os.environ.get("SECRET_KEY",                           226
    "mine-restoration-dev-key")                                         227
    # 单机部署，SQLite 数据库文件放在应用目录下                         228
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                            229
    os.path.join(BASE_DIR, "app.db")                                    230
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              231
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项                  232
    WTF_CSRF_ENABLED = True                                             233
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录 234
    REMEMBER_COOKIE_DURATION = timedelta(days=30)                       235
    REMEMBER_COOKIE_HTTPONLY = True                                     236
    REMEMBER_COOKIE_SAMESITE = "Lax"                                    237
    # 影像与数据文件上传（遥感影像挂接、批量导入）                     238
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",                    239
    "uploads")                                                          240
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限       241
    32 MB                                                               242
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",                   243
    "webp", "bmp", "tif", "tiff"}                                       244
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}                           245
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积 246
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",                  247
    "webp"}                                                             248
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024             # 4 MB              249
# ==== app/models.py ====                                             250
```

```
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」中自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 显示名称（姓名）
    title = db.Column(db.String(64), nullable=True)
    # 职务 / 岗位
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def shown_name(self):
        """界面上显示的名字：优先用显示名称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字母占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
```

```
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
```

```
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)   # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)   # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)     # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
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    __tablename__ = "standard_rules"                                    401
    id = db.Column(db.Integer, primary_key=True)                        402
    standard_id = db.Column(db.Integer,                                 403
    db.ForeignKey("standards.id"), nullable=False)                      404
    item = db.Column(db.String(128), nullable=False)                    405
    # 判定项                                                            406
    operator = db.Column(db.String(8), nullable=False)                  407
    # 比较符：> >= < <= ==                                              408
    threshold = db.Column(db.String(64), nullable=True)                 409
    # 阈值                                                              410
    condition_type = db.Column(db.String(32),                           411
    default="必要条件")  # 必要条件 / 充分条件                          412
    region = db.Column(db.String(64), nullable=True)                    413
    # 适用分区（宁夏分区差异化）                                       414
    note = db.Column(db.Text, nullable=True)                            415
class ConstraintCheck(db.Model):                                        416
    """限制性判定记录。"""                                              417
    __tablename__ = "constraint_checks"                                 418
    id = db.Column(db.Integer, primary_key=True)                        419
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            420
    ches.id"), nullable=False)                                          421
    policy_pass = db.Column(db.Boolean, nullable=True)                  422
    # 政策约束                                                          423
    location_pass = db.Column(db.Boolean, nullable=True)                424
    # 区位约束                                                          425
    pollution_pass = db.Column(db.Boolean, nullable=True)               426
    # 土壤污染                                                          427
    geo_pass = db.Column(db.Boolean, nullable=True)                     428
    # 地质灾害                                                          429
    farmland_pass = db.Column(db.Boolean, nullable=True)                430
    # 耕地属性                                                          431
    passed = db.Column(db.Boolean, nullable=True)                       432
    # 总体是否通过                                                      433
    reason = db.Column(db.Text, nullable=True)                          434
    # 不通过原因                                                        435
    checked_at = db.Column(db.DateTime,                                 436
    default=datetime.utcnow)                                            437
class MicroUnit(db.Model):                                              438
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                      439
    __tablename__ = "micro_units"                                       440
    id = db.Column(db.Integer, primary_key=True)                        441
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            442
    ches.id"), nullable=False)                                          443
    unit_type = db.Column(db.String(32), nullable=False)                444
    # 单元类型                                                          445
    area = db.Column(db.Float, nullable=True)                           446
    # 面积                                                              447
    ratio = db.Column(db.Float, nullable=True)                          448
    # 面积占比（自动计算）                                              449
    description = db.Column(db.Text, nullable=True)                     450
```

```
    #  现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)   #  指标编码
    name = db.Column(db.String(64), nullable=False)
    #  指标名称
    level = db.Column(db.Integer, nullable=False)
    #  1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    #  父指标编码
    #  角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    #  单位
    ideal_value = db.Column(db.String(64), nullable=True)
    #  理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    #  benefit/cost/mid/discrete
    #  启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    #  取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    #  所属单元类型
    name = db.Column(db.String(64), nullable=True)
    #  方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)         #  1=一级指标  2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    #  是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
```

```
    nullable=False)        #  指标编码                                    501
    weight = db.Column(db.Float, nullable=False)                          502
    #  权重值                                                             503
class ExpertMatrix(db.Model):                                             504
    """专家判断矩阵：多专家打分的原始矩阵。"""                              505
    __tablename__ = "expert_matrices"                                     506
    id = db.Column(db.Integer, primary_key=True)                          507
    scheme_id = db.Column(db.Integer,                                     508
    db.ForeignKey("weight_schemes.id"), nullable=True)                    509
    #  直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             510
    unit_type = db.Column(db.String(32), nullable=True)                   511
    #  微地质单元类型                                                     512
    l1_name = db.Column(db.String(64), nullable=True)                     513
    #  所属一级指标（二级矩阵用）                                         514
    expert_name = db.Column(db.String(64), nullable=True)                 515
    #  专家姓名                                                           516
    level = db.Column(db.Integer, nullable=False,                         517
    default=1)          #  矩阵对应指标层级                               518
    matrix_json = db.Column(db.Text, nullable=False)                      519
    #  判断矩阵（ＪＳＯＮ 二维数组）                                      520
    cr = db.Column(db.Float, nullable=True)                               521
    #  该专家矩阵的一致性比例                                             522
    created_at = db.Column(db.DateTime,                                   523
    default=datetime.utcnow)                                              524
class IndicatorData(db.Model):                                            525
    """指标实测数据：按微地质单元组织。"""                                  526
    __tablename__ = "indicator_data"                                      527
    id = db.Column(db.Integer, primary_key=True)                          528
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni              529
    ts.id"), nullable=False, unique=True)                                 530
    soil_ratio = db.Column(db.Float, nullable=True)                       531
    #  土石比                                                             532
    thickness = db.Column(db.Float, nullable=True)                        533
    #  土层厚度  cm                                                       534
    ph = db.Column(db.Float, nullable=True)                               535
    organic_matter = db.Column(db.Float, nullable=True)                   536
    #  有机质  g/kg                                                       537
    #  重金属含量，ＪＳＯＮ 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null 538
    heavy_metals = db.Column(db.Text, nullable=True)                      539
    coverage = db.Column(db.Float, nullable=True)                         540
    #  植被覆盖度  %                                                      541
    shannon = db.Column(db.Float, nullable=True)                          542
    #  Shannon-Wiener 多样性指数                                          543
    slope = db.Column(db.Float, nullable=True)                            544
    #  坡度  °                                                            545
    aspect = db.Column(db.String(16), nullable=True)                      546
    #  坡向：阳坡/阴阳坡/阴坡                                             547
    def heavy_metals_dict(self):                                          548
        return json.loads(self.heavy_metals) if                           549
    self.heavy_metals else {}                                             550
```

```
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator，
```

```
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
class ProfileForm(FlaskForm):
    """个人资料：显示名称、职务与联系方式。字段都可留空。"""
    display_name = StringField("显示名称",
    validators=[Optional(), Length(max=64)])
    title = StringField("职务", validators=[Optional(),
    Length(max=64)])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
```

```
    old_password = PasswordField("当前密码",                                    651
    validators=[DataRequired()])                                                652
    new_password = PasswordField("新密码",                                      653
                                        validators=[DataRequired(),             654
    Length(min=6, max=64,                                                       655
                                                                                656
    message="新密码至少 6 位")])                                                657
    confirm = PasswordField("确认新密码",                                       658
                                    validators=[DataRequired(),                 659
                                                                                660
    EqualTo("new_password",                                                     661
                                                                                662
    message="两次输入的新密码不一致")])                                         663
    submit_pwd = SubmitField("修改密码")                                        664
# ==== app/utils/constants.py ====                                             665
# -*- coding: utf-8 -*-                                                        666
"""评价指标常量表。                                                            667
全部数值取自论文，作为系统计算的唯一基准：                                     668
- 论文 4.2.2 节：指标标准化理想极值与取值依据                                  669
- 论文 4.2.3 节 / 表 4：适宜性等级划分                                         670
- 论文表 3：按微地质单元分别确定的一级、二级指标权重                           671
- 论文 4.2.1 节：平均随机一致性指标 RI                                         672
"""                                                                            673
# ------------------------------------------------------------------------     674
    ----------------                                                           675
# 一、微地质单元类型                                                           676
# ------------------------------------------------------------------------     677
    ----------------                                                           678
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡           679
# 三种地质微单元类型"                                                          680
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")                                  681
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）  682
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"                         683
UNIT_ENABLED_INDICATORS = {                                                    684
    "岩质边坡": ("植被指标", "地形指标"),              # 不设土壤指标            685
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),                          686
    "平台": ("土壤指标", "植被指标"),                  # 不设地形指标            687
}                                                                              688
# ------------------------------------------------------------------------     689
    ----------------                                                           690
# 二、指标标准化理想极值（论文 4.2.2 节）                                      691
# ------------------------------------------------------------------------     692
    ----------------                                                           693
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值                                 694
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）                        695
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）                    696
IDEAL_ORGANIC = 30.0            # 有机质理想值                                 697
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值                     698
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）                 699
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统        700
```

```
#  重金属极值 （mg/kg），《绿化种植土壤》ＣＪ/Ｔ ３４０—２０１６ 土壤重金属含量 Ⅳ 级最高值
#  论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
#  污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
#  坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
#  坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
#  pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
#  ------------------------------------------------------------------------
    ------------------
#  三、适宜性等级划分（论文表 4）
#  ------------------------------------------------------------------------
    ------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    #  score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
#  ------------------------------------------------------------------------
    ------------------
#  四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2，3，4，5，其余为常用扩展值）
```

```
#  ----------------------------------------------------------------------  751
      ----------------                                                      752
RI_TABLE  =  {1:  0.0,  2:  0.0,  3:  0.58,  4:  0.90,  5:  1.12,  6:      753
      1.24,                                                                 754
                  7:  1.32,  8:  1.41,  9:  1.45,  10:  1.49}               755
#  一致性比例阈值（论文：当  CR<0.1  时，判断矩阵满足一致性要求）           756
CR_THRESHOLD  =  0.1                                                        757
#  ----------------------------------------------------------------------  758
      ----------------                                                      759
#  五、默认权重方案（论文表  3）                                           760
#  ----------------------------------------------------------------------  761
      ----------------                                                      762
#  论文  4.2.2  节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  763
#  分别计算评价指标权重"（论文摘要、图  2  流程与表  5  均作"碎石边坡"，      764
#  系统统一采用"碎石边坡"）。权重由  10  位专家打分后经  AHP  方法确定。      765
#  下表为论文表  3  所载的归一化权重，作为系统默认方案。                    766
DEFAULT_WEIGHTS  =  {                                                       767
      "岩质边坡":  {                                                        768
            "l1":  {"植被指标":  0.45,  "地形指标":  0.55},                 769
            "l2":  {                                                        770
                  "植被指标":  {"植被覆盖度":  0.67,  "生物多样性":  0.33},  771
                  "地形指标":  {"坡度":  0.64,  "坡向":  0.36},             772
            },                                                              773
      },                                                                    774
      "碎石边坡":  {                                                        775
            "l1":  {"土壤指标":  0.44,  "植被指标":  0.31,  "地形指标":  0.25},  776
            "l2":  {                                                        777
                  "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,  778
                              "有机质":  0.17,  "污染指标":  0.17},         779
                  "植被指标":  {"植被覆盖度":  0.64,  "生物多样性":  0.36},  780
                  "地形指标":  {"坡度":  0.67,  "坡向":  0.33},             781
            },                                                              782
      },                                                                    783
      "平台":  {                                                            784
            "l1":  {"土壤指标":  0.54,  "植被指标":  0.46},                 785
            "l2":  {                                                        786
                  "土壤指标":  {"土石比":  0.25,  "土层厚度":  0.25,  "pH":  0.17,  787
                              "有机质":  0.17,  "污染指标":  0.17},         788
                  "植被指标":  {"植被覆盖度":  0.61,  "生物多样性":  0.39},  789
            },                                                              790
      },                                                                    791
}                                                                           792
#  ----------------------------------------------------------------------  793
      ----------------                                                      794
#  六、限制性指标（论文表  2  +  4.1  节流程）                             795
#  ----------------------------------------------------------------------  796
      ----------------                                                      797
#  论文摘要与  4.1  节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  798
CONSTRAINT_ITEMS  =  ("政策",  "区位约束",  "土壤污染",  "地质灾害",  "耕地属性")  799
#  ----------------------------------------------------------------------  800
```

```
    ------------------                                                  801
# 七、用户角色                                                          802
# ------------------------------------------------------------------- 803
    ------------------                                                  804
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",                 805
    "viewer": "查看者"}                                                 806
# ==== app/utils/schema.py ====                                       807
# -*- coding: utf-8 -*-                                               808
"""轻量结构迁移。                                                       809
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。            810
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，             811
使旧数据库无需重建即可用上新字段。                                     812
"""                                                                     813
from sqlalchemy import text                                             814
# (表名, 列名, 列定义)                                                  815
REQUIRED_COLUMNS = [                                                    816
    # 遥感影像挂接                                                      817
    ("mine_patches", "image_path", "VARCHAR(256)"),                     818
    ("mine_patches", "image_note", "VARCHAR(256)"),                     819
    ("mine_patches", "image_uploaded_at", "DATETIME"),                  820
    # 指标维护                                                          821
    ("indicators", "enabled_units", "VARCHAR(128)"),                    822
    ("indicators", "description", "TEXT"),                              823
    # 多专家矩阵                                                        824
    ("expert_matrices", "unit_type", "VARCHAR(32)"),                    825
    ("expert_matrices", "l1_name", "VARCHAR(64)"),                      826
    ("expert_matrices", "cr", "FLOAT"),                                 827
    # 用户个人资料（个人设置页）                                        828
    ("users", "display_name", "VARCHAR(64)"),                           829
    ("users", "title", "VARCHAR(64)"),                                  830
    ("users", "email", "VARCHAR(128)"),                                 831
    ("users", "phone", "VARCHAR(32)"),                                  832
    ("users", "avatar", "VARCHAR(256)"),                                833
]                                                                       834
def ensure_schema(db):                                                  835
    """为已有表补齐缺失列。返回补列清单。"""                           836
    added = []                                                          837
    try:                                                                838
        with db.engine.connect() as conn:                               839
            for table, column, ddl in REQUIRED_COLUMNS:                 840
                rows = conn.execute(text("PRAGMA                        841
    table_info(%s)" % table)).fetchall()                                842
                if not rows:                                            843
                    continue                                #           844
    表还不存在，create_all 会建                                         845
                existing = {r[1] for r in rows}                         846
                if column in existing:                                  847
                    continue                                            848
                conn.execute(text(                                      849
                    "ALTER TABLE %s ADD COLUMN %s %s" %                 850
```

```
        (table, column, ddl))                                                  851
                    conn.commit()                                              852
                    added.append("%s.%s" % (table, column))                    853
        except Exception as e:                                  #              854
        迁移失败不应阻断启动                                                   855
            print("[schema] 结构检查跳过：%s" % e)                             856
        return added                                                           857
# ==== app/utils/uploads.py ====                                               858
# -*- coding: utf-8 -*-                                                        859
"""上传文件处理（遥感影像、批量导入）。"""                                     860
import os                                                                      861
import re                                                                      862
import uuid                                                                    863
from datetime import datetime                                                  864
from werkzeug.utils import secure_filename                                     865
def _safe_ext(filename):                                                       866
        return os.path.splitext(filename)[1].lower().lstrip(".")               867
def ext_allowed(filename, allowed):                                            868
        return _safe_ext(filename) in allowed                                  869
def save_upload(file_storage, subdir, allowed_ext,                             870
        upload_root):                                                          871
        """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。  872
        文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。        873
        """                                                                    874
        if file_storage is None or not file_storage.filename:                  875
            return None, "未选择文件"                                          876
        ext = _safe_ext(file_storage.filename)                                 877
        if ext not in allowed_ext:                                             878
            return None, "不支持的文件类型：.%s" % ext                         879
        target_dir = os.path.join(upload_root, subdir)                         880
        os.makedirs(target_dir, exist_ok=True)                                 881
        # 仅保留安全字符，并加时间戳与随机串防重名                             882
        base = secure_filename(os.path.splitext(file_storage.fil               883
        ename)[0]) or "file"                                                   884
        base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                    885
        "file"                                                                 886
        name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M               887
        %S"),                                                                  888
                             uuid.uuid4().hex[:6], ext)                        889
        abs_path = os.path.join(target_dir, name)                              890
        file_storage.save(abs_path)                                            891
        rel = os.path.join("uploads", subdir,                                  892
        name).replace("\\", "/")                                               893
        return rel, None                                                       894
def remove_upload(rel_path, upload_root):                                      895
        """删除已保存的上传文件（相对 static 的路径）。"""                     896
        if not rel_path:                                                       897
            return                                                             898
        abs_path = os.path.join(upload_root,                                   899
        rel_path.replace("uploads/", "", 1))                                   900
```

```
    try:                                                                    901
        if os.path.isfile(abs_path):                                        902
            os.remove(abs_path)                                             903
    except OSError:                                                         904
        pass                                                                905
# ==== app/utils/charts.py ====                                             906
# -*- coding: utf-8 -*-                                                     907
"""轻量图表。                                                               908
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                                 909
- 服务端渲染，页面与打印（PDF）中都能稳定显示                               910
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）                  911
- 单色方案，符合界面「一个视图一个强调色」的约定                            912
"""                                                                         913
from html import escape                                                     914
ACCENT = "#1f3a5f"                                                          915
ACCENT_SOFT = "#7f93ad"                                                     916
INK = "#23262b"                                                             917
MUTED = "#6b7280"                                                           918
LINE = "#dfe3e8"                                                            919
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                              920
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                        921
    "#bcc9da", "#dbe3ec"]                                                   922
def _fmt(v, digits=4):                                                      923
    try:                                                                    924
        return ("%%.%df" % digits) % float(v)                               925
    except (TypeError, ValueError):                                         926
        return str(v)                                                       927
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,                  928
    value_w=76):                                                            929
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                       930
    if not items:                                                           931
        return '<p class="muted">暂无数据</p>'                              932
    n = len(items)                                                          933
    height = n * (bar_h + gap) + 10                                         934
    max_v = max((abs(float(v)) for _, v, _ in items),                       935
    default=1) or 1                                                         936
    body_w = width - label_w - value_w                                      937
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                        938
    height="%d" '                                                           939
             'role="img" aria-label="条形图" '                              940
             'style="max-width:%dpx;font-family:inherit;">'                 941
    % (width, height, height, width)]                                       942
    for i, (label, value, disp) in enumerate(items):                        943
        y = i * (bar_h + gap) + 5                                           944
        frac = (abs(float(value)) / max_v) if max_v else 0                  945
        w = max(2.0, body_w * frac)                                         946
        fill = GRAYS[i % len(GRAYS)]                                        947
        parts.append(                                                       948
            '<text x="%d" y="%d" text-anchor="end"                          949
    font-size="12" fill="%s">%s</text>'                                     950
```

```
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
            'style="font-variant-numeric:tabular-nums;">%s</
    text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
    escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, float(v), d) for l, v, d in items if
    float(v) > 0]
    if not vals:
        return '<p class="muted">暂无数据</p>'
    total = sum(v for _, v, _ in vals) or 1
    r = (size - thickness) / 2
    cx = cy = size / 2
    import math
    parts = ['<svg viewBox="0 0 %d %d" width="%d"
    height="%d" role="img" '
             'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
            % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
```

```
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
            % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
               100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"
    ,
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom
    e.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed
    ge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe
    ",
]
#  允许通过环境变量覆盖，便于在其他机器上指定浏览器
```

```
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
```

```
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
```

```
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
```

```
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式（8） 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
```

```
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
    IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
```

```
                                std_benefit(data.get("thickness"),    1301
    IDEAL_THICKNESS),                                                 1302
                                l2w["土壤指标"]["土层厚度"]),                1303
            "pH": (data.get("ph"),                                    1304
                   std_mid_ph(data.get("ph"),                         1305
    *IDEAL_PH_RANGE),                                                 1306
                   l2w["土壤指标"]["pH"]),                               1307
            "有机质": (data.get("organic_matter"),                     1308
                       std_benefit(data.get("organic_matter")         1309
    , IDEAL_ORGANIC),                                                 1310
                       l2w["土壤指标"]["有机质"]),                          1311
            "污染指标": (data.get("heavy_metals") or {},               1312
                         pollution_score(data.get("heavy_meta         1313
    ls") or {}),                                                      1314
                         l2w["土壤指标"]["污染指标"]),                        1315
        }                                                             1316
    # ---- 植被指标 ----                                              1317
    if "植被指标" in l1_enabled:                                        1318
        result["植被指标"] = {                                          1319
            "植被覆盖度": (data.get("coverage"),                        1320
                           std_benefit(data.get("coverage"),          1321
    IDEAL_COVERAGE),                                                  1322
                           l2w["植被指标"]["植被覆盖度"]),                    1323
            "生物多样性": (data.get("shannon"),                         1324
                           std_benefit(data.get("shannon"),           1325
    IDEAL_SHANNON),                                                   1326
                           l2w["植被指标"]["生物多样性"]),                    1327
        }                                                             1328
    # ---- 地形指标 ----                                              1329
    if "地形指标" in l1_enabled:                                        1330
        result["地形指标"] = {                                          1331
            "坡度": (data.get("slope"),                               1332
                     std_slope(data.get("slope")),                    1333
                     l2w["地形指标"]["坡度"]),                            1334
            "坡向": (data.get("aspect"),                              1335
                     aspect_score(data.get("aspect")),                1336
                     l2w["地形指标"]["坡向"]),                            1337
        }                                                             1338
    return result                                                     1339
# ==== app/services/weight_service.py ====                          1340
# -*- coding: utf-8 -*-                                             1341
"""权重方案的存取与生效权重解析。                                          1342
系统有两套权重来源：                                                    1343
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），       1344
   是**权威取值来源**，案例复算与报告结论以此为准；                          1345
2. 通过层次分析法在本系统中实际算出的权重，保存在                           1346
   weight_schemes / weight_items 两张表中，可标记为「生效」。              1347
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                      1348
以论文表 3 为底，用生效方案中已保存的条目覆盖。                             1349
没有任何生效方案时，即完全使用论文表 3 的权重。                             1350
```

```
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
# 方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights       嵌套权重结构，可直接交给标准化与评价引擎
        source_label  权重来源说明，用于界面与报告标注
        scheme        生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
```

```
    merged["l2"][l1_name]:                                          1401
                merged["l2"][l1_name][l2_name] = w                  1402
                applied += 1                                        1403
    if applied == 0:                                                1404
        return copy.deepcopy(base), "论文表 3 默认权重", None          1405
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme   1406
def save_weights(unit_type, code_weight_pairs, note=None):          1407
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。          1408
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）              1409
    分次录入而无需一次填完。                                              1410
    """                                                             1411
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1412
    e,                                                              1413
                                                                    1414
    name=ACTIVE_NAME).first()                                       1415
    if scheme is None:                                              1416
        scheme = WeightScheme(unit_type=unit_type,                  1417
    name=ACTIVE_NAME,                                               1418
                                  level=1, is_default=True)         1419
        db.session.add(scheme)                                      1420
        db.session.flush()                                          1421
    if note:                                                        1422
        scheme.level = scheme.level or 1                            1423
    if not scheme.is_default:                                       1424
        scheme.is_default = True                                    1425
    written = 0                                                     1426
    for code, w in code_weight_pairs:                               1427
        if w is None or w <= 0:                                     1428
            continue                                                1429
        item = WeightItem.query.filter_by(scheme_id=scheme.i        1430
    d,                                                              1431
                                                                    1432
    indicator_code=code).first()                                    1433
        if item is None:                                            1434
            item = WeightItem(scheme_id=scheme.id,                  1435
    indicator_code=code)                                            1436
            db.session.add(item)                                    1437
        item.weight = float(w)                                      1438
        written += 1                                                1439
    db.session.commit()                                             1440
    return scheme, written                                          1441
def reset_weights(unit_type):                                       1442
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""                1443
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1444
    pe,                                                             1445
                                                                    1446
    name=ACTIVE_NAME).all()                                         1447
    n = 0                                                           1448
    for s in schemes:                                               1449
        WeightItem.query.filter_by(scheme_id=s.id).delete(          1450
```

```
                synchronize_session=False)
            db.session.delete(s)
            n += 1
    db.session.commit()
    return n
def scheme_items(unit_type):
    """返回该单元类型已保存的自定义权重条目 {指标编码: 权重}。"""
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return {}, None
    items = {i.indicator_code: i.weight
             for i in WeightItem.query.filter_by(scheme_id=scheme.id).all()}
    return items, scheme
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type)
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式(9) 实现：
- 单元级：F' = Σ(二级指标权重 × 该二级指标下三级指标标准化值)
- 矿山级：F = Σ(Si × F'i)，Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details: [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
```

```
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ------------------------------------------------------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
```

```
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】 （面积占比 %.2f%%）" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
```

```
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码:", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码:", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）:", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
```

```
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3　提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 （0.44, 0.31, 0.25） 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 = 0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4　不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
```

```
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
# 工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据,避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
```

```
            db.session.delete(p)
        db.session.commit()
print("    已清理测试项目")
print()
print(LINE)
print("测试 4    微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
        patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
        irst()
        pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("    单元列表含三类单元：", all(t in body for t in ("岩质边坡",
        "碎石边坡", "平台")))
print("    含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5    标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("    含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
        "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("    口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("    含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6    限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    含五项条件：", all(x in body for x in ("政策约束", "区位约束",
        "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
        "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "1",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
```

```
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
```

```
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
```

```
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.id.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
```

```
        rec = save_evaluation(pid, units)
        rid = rec.id
        print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
              % (rid, rec.final_score, rec.grade))
        from app.models import ScoreDetail
        n_detail = ScoreDetail.query.filter_by(result_id=rid).co
        unt()
        print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
      ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
      "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
```

```
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
```

```
      "[不符合]"))                                                    10533
# 2) 勾选后应下发带过期时间的 remember_token                          10534
c = app.test_client()                                                 10535
r = c.post("/login", data={"username": "admin", "password":           10536
      "admin123",                                                     10537
                                  "remember": "y"},                   10538
      follow_redirects=False)                                         10539
cookies = r.headers.getlist("Set-Cookie")                             10540
rt = [h for h in cookies if h.startswith("remember_token=")]          10541
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else          10542
      "[不符合]"))                                                    10543
assert rt, "未下发 remember_token"                                    10544
exp = re.search(r"Expires=([^;]+)", rt[0])                            10545
print("  %s 该 cookie 带过期时间：%s"                                  10546
        % ("[符合]" if exp else "[不符合]", exp.group(1) if exp        10547
      else "无（会话 cookie）"))                                      10548
assert exp, "remember_token 无过期时间，关浏览器即失效"               10549
# 3) 不勾选时不应下发 remember_token                                  10550
c2 = app.test_client()                                                10551
r2 = c2.post("/login", data={"username": "admin",                     10552
      "password": "admin123"},                                        10553
                follow_redirects=False)                               10554
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if                 10555
      h.startswith("remember_token=")]                                10556
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2      10557
      else "[不符合]"))                                               10558
assert not rt2                                                        10559
# 4) 携带 remember_token 的新会话应自动登录                           10560
c3 = app.test_client()                                                10561
c3.set_cookie("remember_token",                                       10562
                rt[0].split(";")[0].split("=", 1)[1],                 10563
      domain="localhost")                                             10564
r3 = c3.get("/", follow_redirects=False)                              10565
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"     10566
        % ("[符合]" if r3.status_code == 200 else "[不符合]",         10567
      r3.status_code))                                                10568
assert r3.status_code == 200                                          10569
# ------------------------------------------------------------------  10570
      ----------------                                                10571
print()                                                               10572
print(LINE)                                                           10573
print("二、删除功能")                                                 10574
print(LINE)                                                           10575
client.post("/login", data={"username": "admin",                      10576
      "password": "admin123"})                                        10577
# 项目：列表页须有删除按钮                                            10578
html = client.get("/project/").get_data(as_text=True)                 10579
has_del = "/delete" in html and "btn--danger" in html                 10580
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))  10581
assert has_del, "项目列表缺少删除按钮"                                10582
```

```
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else "[不符合]"))
assert "confirm(" in html
# 图斑列表
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
html = client.get("/project/%d/patches" % patch_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in html and "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=True)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）" % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目", "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id, follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
```

```
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
```

```
        "admin123"})
def ok(flag, label, extra=""):
        print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
        label, extra))
        assert flag, label
# ------------------------------------------------------------------
        ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
        before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
        "LOG-T"})
with app.app_context():
        after = OperationLog.query.count()
        last = OperationLog.query.order_by(OperationLog.id.desc(
        )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
        after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
        last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
        u = User(username="tmp_user", role="viewer",
                 password_hash=User.query.first().password_hash)
        db.session.add(u); db.session.commit()
        uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
        "active": "1"})
with app.app_context():
        ok(db.session.get(User, uid).role == "evaluator",
        "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
        admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
        "active": "1"})
with app.app_context():
        ok(db.session.get(User, admin_id).role == "admin",
        "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
```

```
with app.app_context():
    ok(db.session.get(User, uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
```

```
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------------
    ----------------
```

```
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ----------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
```

```
        "０.４０００")])                                                  10883
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                  10884
#  报告是评价结论的呈现，需先真实执行一次评价才能打开                      10885
with app.app_context():                                                  10886
    from app.views.evaluate import load_patch_input,                     10887
    save_evaluation                                                      10888
    save_evaluation(patch_id, load_patch_input(patch_id))                10889
r = c.get("/report/%d" % patch_id)                                       10890
body = r.get_data(as_text=True)                                          10891
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %            10892
    body.count("<svg"))                                                  10893
exe = browser.browser_path()                                             10894
if exe is None:                                                          10895
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")                       10896
else:                                                                    10897
    r = c.get("/report/%d/pdf" % patch_id)                               10898
    data = r.data                                                        10899
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %              10900
    len(data))                                                           10901
    d = fitz.open(stream=data, filetype="pdf")                           10902
    txt = "".join(d[i].get_text() for i in                               10903
    range(d.page_count))                                                 10904
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")                 10905
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")   10906
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)        10907
    d.close()                                                            10908
print()                                                                  10909
print(LINE)                                                              10910
print("补齐模块自检全部通过 √")                                          10911
print(LINE)                                                              10912
#  ==== 测试/test_workflow.py ====                                       10913
#  -*- coding: utf-8 -*-                                                 10914
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                  10915
论文 ４.１ 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。  10916
本测试逐个构造「缺前置」的场景，验证系统不会放行。                      10917
"""                                                                      10918
import os                                                                10919
import sys                                                               10920
sys.stdout.reconfigure(encoding="utf-8")                                 10921
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，             10922
#  使 `from app import ...` 与 app/app.db 的定位不受影响。               10923
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil             10924
    e__)))                                                               10925
if ROOT not in sys.path:                                                 10926
    sys.path.insert(0, ROOT)                                             10927
from app import create_app                                               10928
from app.models import (db, Project, MinePatch, MicroUnit,               10929
    IndicatorData,                                                       10930
                            ConstraintCheck, EvaluationResult)           10931
from app.services import workflow as wf                                  10932
```

```
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
```

```
                                                ph=9.24,                    10983
        organic_matter=4.89, coverage=87.92,                                10984
                                                shannon=0.60,               10985
        slope=28.75, aspect="阳坡"))                                        10986
            db.session.commit()                                             10987
def cleanup(pid, tag):                                                      10988
        with app.app_context():                                             10989
            from app.models import ScoreDetail                              10990
            #  先删得分明细，再删评价记录，避免留下孤儿明细                10991
            rids = [r.id for r in EvaluationResult.query.filter_            10992
        by(patch_id=pid)]                                                   10993
            if rids:                                                        10994
                ScoreDetail.query.filter(ScoreDetail.result_id.i            10995
        n_(rids)).delete(                                                   10996
                    synchronize_session=False)                              10997
            EvaluationResult.query.filter_by(patch_id=pid).delet            10998
        e()                                                                 10999
            ConstraintCheck.query.filter_by(patch_id=pid).delete            11000
        ()                                                                  11001
            uids = [u.id for u in MicroUnit.query.filter_by(pat             11002
        h_id=pid)]                                                          11003
            if uids:                                                        11004
                IndicatorData.query.filter(IndicatorData.unit_id            11005
        .in_(uids)).delete(                                                 11006
                    synchronize_session=False)                              11007
            MicroUnit.query.filter_by(patch_id=pid).delete()                11008
            m = db.session.get(MinePatch, pid)                              11009
            if m:                                                           11010
                db.session.delete(m)                                        11011
            pr = Project.query.filter_by(name="门禁-" +                     11012
        tag).first()                                                        11013
            if pr:                                                          11014
                db.session.delete(pr)                                       11015
            db.session.commit()                                             11016
def results_of(pid):                                                        11017
        with app.app_context():                                             11018
            return EvaluationResult.query.filter_by(patch_id=pid            11019
        ).count()                                                           11020
def post_eval(pid):                                                         11021
        return c.post("/evaluate/%d" % pid, data={},                        11022
        follow_redirects=False)                                             11023
def get_report(pid):                                                        11024
        return c.get("/report/%d" % pid, follow_redirects=False)            11025
# ------------------------------------------------------------------------  11026
        ------------------                                                  11027
print(LINE)                                                                 11028
print("一、前置条件缺失时，评价必须被拦住")                                 11029
print(LINE)                                                                 11030
print()                                                                     11031
print("场景 1：连微地质单元都没划分")                                       11032
```

```
pid = new_patch("1")                                                    11033
r = post_eval(pid)                                                      11034
n = results_of(pid)                                                     11035
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",          11036
    "HTTP %d，结果数 %d" % (r.status_code, n))                           11037
with app.app_context():                                                 11038
    pr = wf.patch_progress(pid)                                         11039
steps = [b["step"] for b in pr["blocking"]]                             11040
ok(not pr["ready"] and "units" in steps,                                11041
    "拦原因包含「划分微地质单元」", "、".join(steps))                    11042
ok(steps == ["constraint", "units"],                                    11043
    "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",            11044
    "、".join(steps))                                                   11045
cleanup(pid, "1")                                                       11046
print()                                                                 11047
print("场景 2：有单元，但未做限制性条件判定")                           11048
pid = new_patch("2"); add_units(pid); add_data(pid)                     11049
r = post_eval(pid)                                                      11050
n = results_of(pid)                                                     11051
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",    11052
    "结果数 %d" % n)                                                    11053
with app.app_context():                                                 11054
    pr = wf.patch_progress(pid)                                         11055
ok(any(b["step"] == "constraint" for b in pr["blocking"]),              11056
    "拦原因指向「执行限制性条件判定」")                                  11057
cleanup(pid, "2")                                                       11058
print()                                                                 11059
print("场景 3：限制性条件判定【未通过】")                               11060
pid = new_patch("3"); add_units(pid); add_data(pid);                    11061
    add_constraint(pid, passed=False)                                   11062
r = post_eval(pid)                                                      11063
n = results_of(pid)                                                     11064
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",       11065
    "结果数 %d" % n)                                                    11066
with app.app_context():                                                 11067
    pr = wf.patch_progress(pid)                                         11068
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")            11069
blocked = [b for b in pr["blocking"] if b["step"] ==                    11070
    "constraint"]                                                       11071
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项   11072
ok(bool(blocked) and "未通过" in blocked[0]["label"],                    11073
    "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else   11074
    "")                                                                 11075
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),         11076
    "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")  11077
cleanup(pid, "3")                                                       11078
print()                                                                 11079
print("场景 4：判定通过，但指标数据全空")                               11080
pid = new_patch("4"); add_units(pid); add_constraint(pid,               11081
    passed=True)                                                        11082
```

```
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "data" for b in pr["blocking"]),
   "拦原因指向「录入指标数据」")
ok(pr["data"]["missing"] == ["碎石边坡"],
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))
cleanup(pid, "4")
print()
print("场景 5：单元面积占比合计不为 100%")
pid = new_patch("5"); add_constraint(pid, passed=True)
with app.app_context():
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,
    ratio=0.5)
    db.session.add(u); db.session.commit()
    db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.3, thickness=10,
                                 ph=8.0,
    organic_matter=5.0, coverage=80.0,
                                 shannon=0.6, slope=20.0,
    aspect="阳坡"))
    db.session.commit()
r = post_eval(pid)
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")
cleanup(pid, "5")
print()
print(LINE)
print("二、前置条件齐备时才放行")
print(LINE)
pid = new_patch("OK"); add_units(pid); add_constraint(pid,
    passed=True); add_data(pid)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")
r = post_eval(pid)
n = results_of(pid)
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)
print("  重定向到:", r.headers.get("Location"))
print()
print(LINE)
print("三、报告不得先于评价产生")
print(LINE)
pid2 = new_patch("R")    # 什么都不做
r = get_report(pid2)
ok(r.status_code == 302 and "/report/" ==
    (r.headers.get("Location") or ""),
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
```

```
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
```

```
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑 - 守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
```

```
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
        follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
# ==== 测试/test_weights.py ====
# -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
```

```
                                ConstraintCheck, EvaluationResult,
    WeightScheme,
                                WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
```

```
                pollution_pass=True, geo_pass=True,
        farmland_pass=True,
                passed=True, reason="全部通过"))
            db.session.commit()
            return patch.id
def cleanup(pid):
        with app.app_context():
            from app.models import ScoreDetail
            rids = [r.id for r in EvaluationResult.query.filter_
        by(patch_id=pid)]
            if rids:
                ScoreDetail.query.filter(ScoreDetail.result_id.i
        n_(rids)).delete(
                    synchronize_session=False)
            EvaluationResult.query.filter_by(patch_id=pid).delet
        e()
            ConstraintCheck.query.filter_by(patch_id=pid).delete
        ()
            uids = [u.id for u in MicroUnit.query.filter_by(pat
        h_id=pid)]
            if uids:
                IndicatorData.query.filter(IndicatorData.unit_id
        .in_(uids)).delete(
                    synchronize_session=False)
            MicroUnit.query.filter_by(patch_id=pid).delete()
            m = db.session.get(MinePatch, pid)
            if m:
                db.session.delete(m)
            p = Project.query.filter_by(code="WEIGHT-%s" %
        TAG).first()
            if p:
                db.session.delete(p)
            db.session.commit()
def run_eval(pid):
        c.post("/evaluate/%d" % pid, data={},
        follow_redirects=False)
        with app.app_context():
            return EvaluationResult.query.filter_by(patch_id=pid
        ) \
                .order_by(EvaluationResult.id.desc()).first().fi
        nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
        ws.reset_weights(UNIT_TYPE)
        w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
        label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
```

```
    "土壤指标 %.2f" % w["l1"]["土壤指标"])                                    11383
pid = build_case()                                                          11384
base_score = run_eval(pid)                                                  11385
print("  默认权重下的评价值 F = %.4f" % base_score)                          11386
print()                                                                     11387
print(LINE)                                                                 11388
print("二、保存 AHP 算出的权重后，评价应改用新权重")                        11389
print(LINE)                                                                 11390
with app.app_context():                                                     11391
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）      11392
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                     11393
    0.20),                                                                  11394
                                ("terrain", 0.10)])                         11395
    n_items = WeightItem.query.count()                                      11396
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)        11397
with app.app_context():                                                     11398
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)                   11399
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",    11400
    label2)                                                                 11401
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",                 11402
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])                                 11403
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")     11404
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                       11405
    "未覆盖的二级权重仍保留论文取值",                                       11406
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                         11407
new_score = run_eval(pid)                                                   11408
print("  新权重下的评价值 F = %.4f" % new_score)                           11409
ok(abs(new_score - base_score) > 1e-4,                                      11410
    "权重变化确实改变了评价结果",                                           11411
    "%.4f → %.4f" % (base_score, new_score))                                11412
print()                                                                     11413
print(LINE)                                                                 11414
print("三、结果应记录所用权重来源")                                         11415
print(LINE)                                                                 11416
import json                                                                 11417
with app.app_context():                                                     11418
    rec = EvaluationResult.query.filter_by(patch_id=pid) \                  11419
        .order_by(EvaluationResult.id.desc()).first()                       11420
    stored = json.loads(rec.unit_scores)                                    11421
ok(any(u.get("weight_source") for u in stored),                             11422
    "评价记录中标注了权重来源",                                             11423
    stored[0].get("weight_source") or "")                                   11424
print()                                                                     11425
print(LINE)                                                                 11426
print("四、恢复默认后应回到论文取值")                                       11427
print(LINE)                                                                 11428
with app.app_context():                                                     11429
    ws.reset_weights(UNIT_TYPE)                                             11430
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)                   11431
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",         11432
```

```
    label3)                                                          11433
back_score = run_eval(pid)                                           11434
print("   恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,          11435
    base_score))                                                     11436
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")        11437
print()                                                              11438
print(LINE)                                                          11439
print("五、权重首页概览")                                             11440
print(LINE)                                                          11441
with app.app_context():                                              11442
    rows = ws.summary()                                              11443
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))  11444
r = c.get("/ahp/")                                                   11445
body = r.get_data(as_text=True)                                      11446
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")    11447
print()                                                              11448
print(LINE)                                                          11449
print("六、一致性不通过的矩阵不得保存生效")                            11450
print(LINE)                                                          11451
with app.app_context():                                              11452
    ws.reset_weights(UNIT_TYPE)                                      11453
n = 3                                                                11454
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限                11455
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]                  11456
data = {"csrf_token": "", "action": "save"}                          11457
for i, j, v in mats:                                                 11458
    data["m_%d_%d" % (i, j)] = v                                     11459
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,           11460
    data=data,                                                       11461
               follow_redirects=True)                                11462
with app.app_context():                                              11463
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)             11464
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)              11465
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")  11466
cleanup(pid)                                                         11467
print()                                                              11468
print(LINE)                                                          11469
if _fails:                                                           11470
    print("存在未通过项：")                                           11471
    for f in _fails:                                                 11472
        print("  -", f)                                             11473
    sys.exit(1)                                                      11474
print("权重生效链路自检全部通过 √")                                   11475
print(LINE)                                                          11476
# ==== app/services/__init__.py ====                                 11477
# -*- coding: utf-8 -*-                                              11478
# ==== app/utils/__init__.py ====                                    11479
# -*- coding: utf-8 -*-                                              11480
# ==== app/views/__init__.py ====                                    11481
# -*- coding: utf-8 -*-                                              11482
```